

AI UX & Data Visualisation Design Principles (CA6002)

Associate Professor Goh Wooi Boon

College of Computing and Data Science
Nanyang Technological University

email: aswbgo@ntu.edu.sg



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Chapter 4.1 – Human-Centred AI

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Human-Centred AI

What is Human-centred AI?

- Human-centred AI (HCAI) refers to the development of AI technologies that prioritises **human needs**, **values** and **capabilities** at the core of its **design** and **operations**.
- The goals of HCAI are to **increase** human self-efficacy, creativity, responsibility, social connectedness, and to **reduce** malicious actors, biased data and flawed software.
- HCAI is important because it ensures that AI systems focus on human **needs** and **values**. When people understand the **value** of the AI system, it **fosters trust** and **acceptance** among users. This trust is essential for **successful integration** and **adoption** of AI systems.

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Human-Centred AI

Traditional AI vs Human-centred AI

- **Traditional AI** emphasises task automation for **efficiency** and productivity. But HCAI prioritises **human needs**, **values** and **capability** and therefore seeks to **augment human capabilities** rather than replace them.

	 Primary Focus	 Purpose	 Design Philosophy	 User Involvement	 Ethical Concerns	 Compliance
Traditional AI	Task automation and efficiency.	Replace human tasks.	Optimized for performance and scalability.	Limited input in design.	Focus on efficiency, less on biases and transparency.	Limited to predefined rules and data.
HCAI	Human needs, values and capabilities.	Augment human capabilities.	Accessible, user-friendly, and ethically aligned.	Active user involvement in design.	Prioritizes privacy, fairness, and transparency.	Learns and adapts from human behaviors and context.

image from
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Rationalism vs Empiricism

- Socrates's **rationalism** relies primarily on **intellectual reasoning** and **logical analysis** as a primary source of knowledge and justification.
- On the other hand, Leonardo Da Vinci's **empiricism** relies on **sensory exploration** of the world to derive knowledge and understanding.
- Both philosophies have their respective values, but each can colour our approach in designing and deploying AI systems very differently.
- Shneiderman's Roomba robot vacuum cleaner example – the **rationalist thinking** is for Roomba to do the job on its own (efficiently and with minimal user interface) compared to the **empiricist thinking** that seeks to give users greater sense of control and anticipation over the process.



[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

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Rationalism vs Empiricism - The difference

- The **rationalists** are guided by their confidence in the perfectibility of rules and strength of formal method of logic. Their AI solutions emphasise **statistical** and **data-driven algorithmic solutions** and they often believe when designed for optimal performance, these systems will exceed human ability in well-defined tasks.
- The **empiricist** believe in the need to go out and **sense the real world** in all its contextual complexity, diversity, uncertainty and always on the lookout for what can **go wrong** and what could be **made better** (e.g. with user feedback).
- Rationalism with its strength in **logical thinking** and **rigour** should be **balanced** with an empirical outlook, which will help AI designers see **other possibilities** and unanticipated behaviour, for example through **observational strategies**.



[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

[4] WBUR – Is 'Google Flu Trends' Prescient Or Wrong? - <https://www.wbur.org/commonhealth/2013/01/13/google-flu-trends-cdc>

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Rationalism vs Empiricism - Implication for design

- AI practitioner sympathetic to **rationalism** tend to favour autonomous designs in which computers operate reliably **without human oversight**.
- A **human-centred, empiricist-driven** strategy recognises that humans must have **meaningful control** of technology and are **clearly responsible** for the outcomes of **their actions**.
- Designs that facilitate human responsibility must allow users to **anticipate** outcomes (e.g provide useful and timely feedback) that are **predictable** and retain **adequate control**.
- Empiricism-oriented HCAI designers focus on the **users**, their needs and behaviour through **observations** in their natural settings and **participatory design**.



[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

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Desirable Attributes of HCAI Systems

- Among many system attributes, Shneiderman highlighted three key ones that are desirable for HCAI systems and useful to guide their design goals. They are:
 - **Reliable** systems produce expected responses when required to do so and such reliability is the result of having sound software engineering practices, which include having audit trails, rigorous testing and explainable outcomes.
 - **Safety** culture amongst various stakeholders is crucial and should guide operational practices, internal reviews and conformance to industry standard.
 - **Trustworthy** systems are trusted by users when they are independently assessed and certified by well-established agencies and organisations.



[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

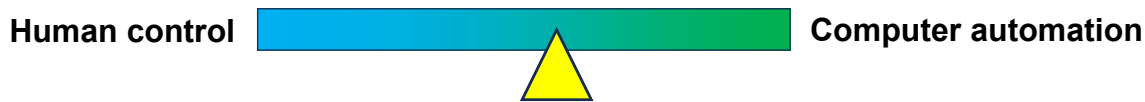
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Human-Centred AI Framework

Human Control vs Computer Automation

- A **one-dimensional view** of **autonomy** believes that an increase in computer automation means a reduction in human control. The desirable goal is often to seek to increase automation and reduce human involvement.



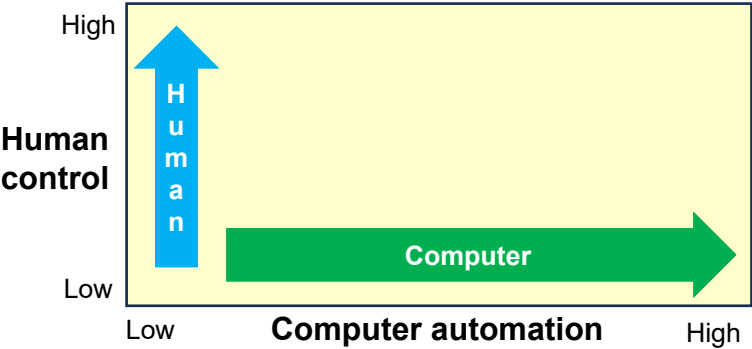
- On the other hand, the **two-dimensional HCAI framework** seeks to decouple from this unnecessary constraint.
- It is possible and desirable to strive to achieve both high levels of human control and high levels of computer automation.

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Human-Centred AI Framework

The 2-Dimensional HCAI Framework

- This two-dimensional framework has two axes, with one going from low to high human control and the other going from low to high computer automation.

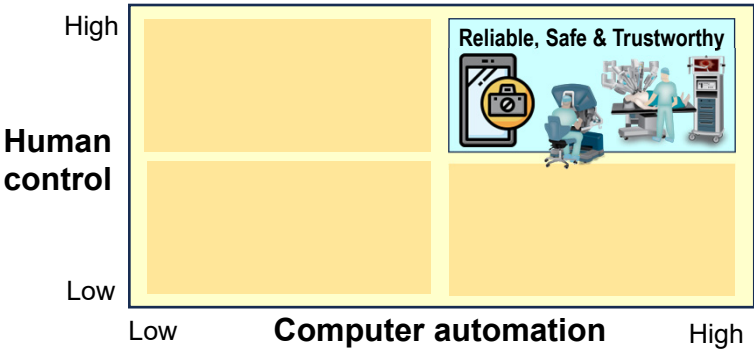


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Human-Centred AI Framework

The 2-Dimensional HCAI Framework

- Poorly understood and **complex** tasks with **varying context of use** fit well into the top right quadrant. Also tasks requiring substantial creative human input.



- Most reliable, safe and trustworthy systems are on this quadrant as they are achieved with high levels of both computer automation and human control.
- Such tasks involve operating cameras, elevators, tele-operated surgical robots, driver-assistance systems, etc.

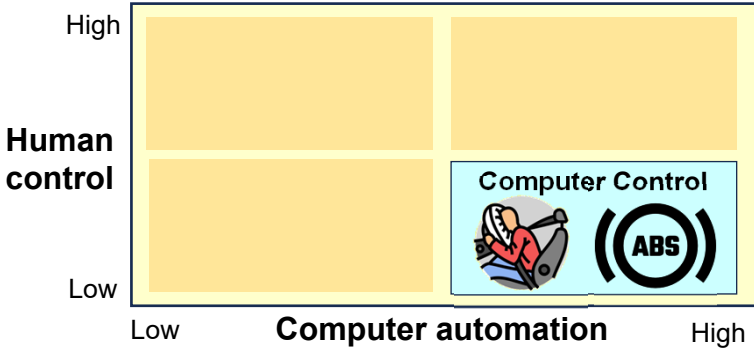


[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

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The 2-Dimensional HCAI Framework

- Lower right quadrant, with **high computer autonomy** and **low human control** is for use cases where **rapid actions** and **control** are beyond human capabilities.



- Such use cases include airbag deployment, anti-lock brake system (ABS), pacemakers, etc.
- Such life critical systems require careful design and testing.
- Systems with highly **stable**, **accurate** and **predictable** processes and sensors can also fall into this quadrant.

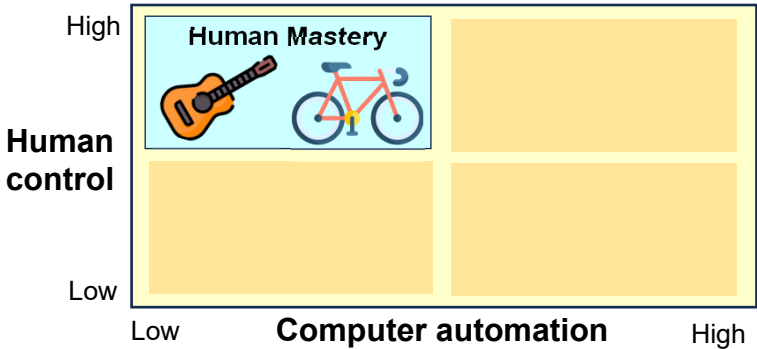


[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

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The 2-Dimensional HCAI Framework

- Top left quadrant are tasks with high human control and low automation, leading to **human mastery**, through free exploration and creative competence building.



- Examples include learning to play the guitar, cycling, baking, leisure sports, etc.
- **Error** is part of the engaging **experience**, which will progressively lead to the mastery of the desired skill and a pleasurable outcome.

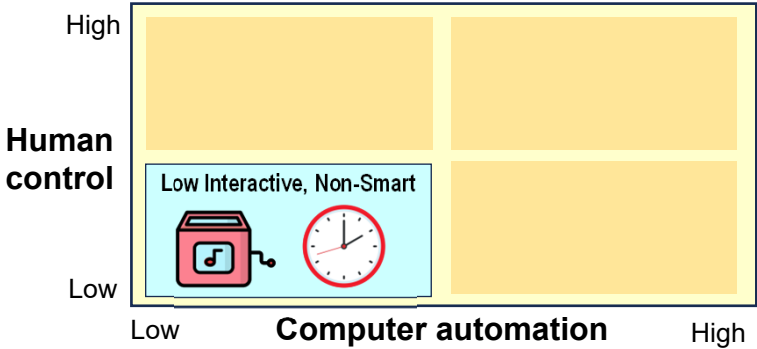


[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

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The 2-Dimensional HCAI Framework

- Lower left quadrant is where devices with **minimal intelligence** and **interactivity** can be found. They have low levels of human control and computer automation.



- Examples of such devices include music boxes, simple clocks (both analog or digital), mousetraps, etc.

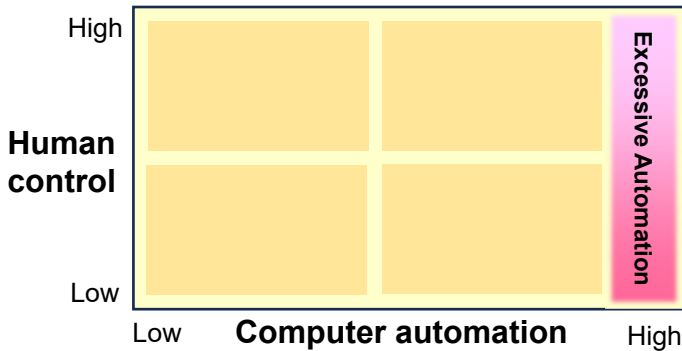


[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

Human-Centred AI Framework

The 2-Dimensional HCAI Framework - Excessive Automation

- The extreme case of full computer autonomy can sometime lead to **deadly outcomes** when designers erroneos believe their autonomous system **cannot fail**.



- This confidence can lead to a lack of consideration for proper manual override or **contingency planning**.
- E.g. the design of the Boeing 737 MAX's MCAS system that led to two tragic plane crashes in 2018 and 2019.
- Pilots were not properly trained on how to switch to manual override when the MCAS malfunctioned.

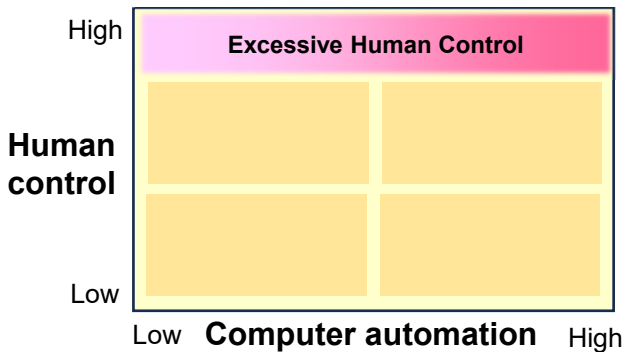


[6] BBC News, Boeing 737 Max MCAS anti-stall system explained - <https://www.bbc.com/news/av/business-47821827>

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The 2-Dimensional HCAI Framework - Excessive Human Control

- Excessive human control of a complex system often leads to **human errors** due to **confusion** of its operation and uncertain **default settings** of the device.



- Life critical systems (e.g. infusion pump) should use **safe** power-up default settings.
- Interlocks** and **guards** help prevent mistakes during unexpected operations (e.g. when microwave door opens during heating cycle).
- Formal methods** and **software-based constraints** (e.g. range checking) ensure algorithms have permissible inputs and produces only acceptable outcomes.

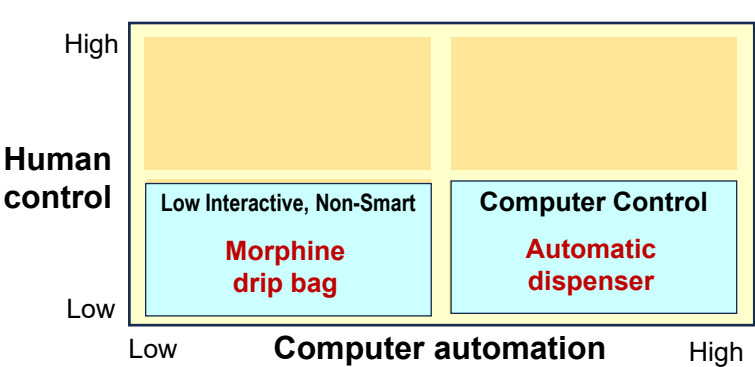


[1] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

Human-Centred AI Framework

The 2-Dimensional HCAI Framework - Case Study

- The four quadrants can help us think about different design possibilities for a product or service. A case study could be the design of an analgesia (i.e. pain killer) delivery system.

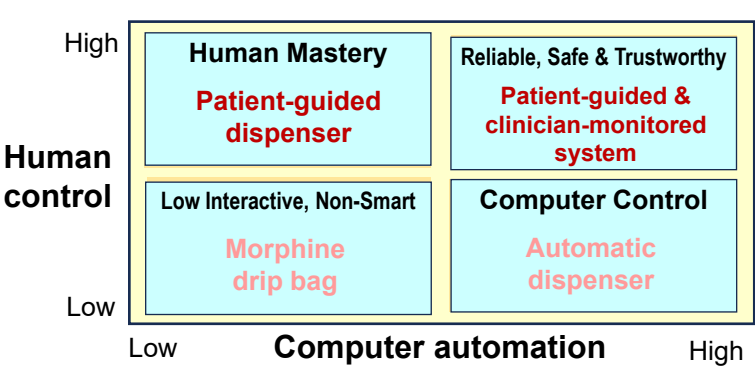


- A low human control and low automation design would be just a simple morphine drip bag.
- An automated system with little human control would provide **pre-programmed** computer selected dosage based on the time of day, patient's activity and vital signs. But with **no assessment** of patient current **sense of pain**.

Human-Centred AI Framework

The 2-Dimensional HCAI Framework - Case Study

- A human-centred design with high human control and low automation will give patients the ability to control the dosage and frequency by squeezing a trigger when in pain.



- Overdosing danger is prevented using **interlocks** that limits maximum frequency and dosage.
- A reliable, safe and trustworthy design **allows users to control** the amount of pain medication but uses **well-tested AI** to **choose dosage** based on patient and disease variables, while preventing overdosing.

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Summary

- **Traditional AI** often justifies its use by automating tasks for increased **efficiency** and **productivity**. But the goals of **HCAI** is to **augment human capability** and **responsibility** rather than replace them.
- HCAI designers should **humbly balance** the **logical rigor** of rationalism with the keen **observational strategies** of empiricism, as real-world systems are **complex** and human behaviour can be **unpredictable** and at times irrational.
- The **2-D HCAI framework** facilitates a view that AI systems can be designed to have both a **high** levels of **human control** and **high** levels of **computer automation**, as such designs often lead to systems with desirable attributes like being **reliable**, **safe** and **trustworthy**.

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- [2] Interaction Design.org, Human-centered AI - <https://www.interaction-design.org/literature/topics/human-centered-ai>
- [3] Hushl.ai, Traditional AI versus Human-centred AI - https://www.linkedin.com/posts/hushl_traditional-ai-vs-human-centric-ai-whats-activity-7220019556201717761-mcnv
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- [6] BBC News, Boeing 737 Max MCAS anti-stall system explained - <https://www.bbc.com/news/av/business-47821827>
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